

18. Perform Edge detection using Sobel Matrix along Y axis

Aim:

To perform edge detection along the Y-axis using the Sobel matrix with the OpenCV library.

Procedure:

- Import the required libraries such as OpenCV (cv2) and NumPy.
- Read the input image using the cv2.imread() function.
- Create the required kernel/parameters for image processing.
- Apply the required image-processing operation using the appropriate OpenCV function.
- Display the original and processed images using cv2.imshow() and close the windows using cv2.destroyAllWindows().

Code:

```
import cv2
```

```
import numpy as np
```

```
cap=cv2.VideoCapture(0)
```

```
while True:
```

```
    ret,frame=cap.read()
```

```
    if not ret:
```

```
        break
```

```
    rows,cols=frame.shape[:2]
```

```
pts1=np.float32([[50,50],[cols-50,50],[50,rows-50],[cols-  
50,rows-50]])  
  
pts2=np.float32([[0,100],[cols,0],[100,rows],[cols-100,rows]])  
  
M=cv2.getPerspectiveTransform(pts1,pts2)  
dst=cv2.warpPerspective(frame,M,(cols,rows))  
  
cv2.imshow("Original",frame)  
cv2.imshow("Perspective Video",dst)  
  
if cv2.waitKey(1)&0xFF==27:  
    break  
  
cap.release()  
cv2.destroyAllWindows()
```

Sample Input



Sample Output



Result:

The edges in the Y-direction were successfully detected using the Sobel Y-axis operator.